Ryan Pratt

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Education

University of Wisconsin–Madison *Expected May 2028*
B.S. in Computer Engineering; Additional Major in Computer Sciences | GPA: 3.73/4.0
Relevant Coursework: Data Structures & Algorithms, Machine Organization & Programming, User Interfaces
In Progress: Operating Systems, High Performance Computing

Skills

Languages: Java, Python, JavaScript, TypeScript, C, C++, SQL, Bash
Frameworks/Libraries: React, Node.js, Express, FastAPI, JUnit, Mockito
Tools/Platforms: Git, Linux, Docker, Maven, SonarQube, Claude Code
Concepts: Data Structures & Algorithms, REST APIs, Unit Testing, Object-Oriented Programming, Full-Stack Development

Experience

Wintrust Financial Corporation – *Full Stack Engineer Intern* *May 2026 – August 2026*
- Increased backend test coverage from 16% to 41% across a roughly 76,000-line Java codebase, writing JUnit, Mockito, and AEM Mocks tests for components, services, and servlets.
- Resolved 400+ SonarQube findings, including critical and major issues, by updating production Java code and deprecated APIs, incorporating peer review feedback, and documenting changes for QA.
- Reduced image payload sizes by approximately 30% by developing a Java OSGi service and integrating WebP delivery into custom Adobe Experience Manager components.

iCode – *Lead Instructor* *June 2025 – August 2025*
- Taught 200+ K–12 students programming and robotics through hands-on projects, explaining technical concepts and guiding debugging.

Technical Experience

Wisconsin Robotics – *Software Team Member* *October 2024 – December 2025*
- Implemented A* path planning in Python for rover navigation, reducing time-to-goal by approximately 15% in simulation and physical testing.
- Integrated navigation software for rover deployment, balancing pathfinding performance with hardware resource constraints.

Wisconsin AI Safety Initiative – *Technical Safety Scholar* *February 2025 – February 2026*
- Analyzed approximately 60 AI research papers and model releases during a 13-week cohort, evaluating model capabilities, robustness, and failure modes.
- Compared alignment and interpretability approaches across academic and open-source systems, communicating technical tradeoffs and potential mitigation strategies in peer discussions.

Projects

Jellyfish – *AI Wiki Search (Badger BuildFest 2026)* *September 2026*
Tech: *Next.js, React, TypeScript, Tailwind CSS, Model Context Protocol (MCP)*
- Won three awards and $2,000 with a four-person team at Badger BuildFest 2026: 2nd place in the Applied AI track, 2nd in the Databricks Real-World Workflows challenge, and a share of 1st in Open Venture.
- Designed the system architecture, a retrieval pipeline that re-ranks wiki search results with a language model, flags pages made stale by an edit, and serves answers to Claude over MCP, and built its Next.js/TypeScript web app.

Azterra – *Full-Stack Interactive Atlas* *September 2025 – December 2025*
Tech: *React, Node.js, Express, Leaflet, Supabase, Netlify*
- Built a full-stack interactive atlas using React and a Node.js/Express REST API, supporting layered maps, custom markers, and filtering across 34K+ custom map tiles.
- Implemented map editing and filtering, multi-provider authentication, role-based access control, and persistence for 25+ datasets.
- Deployed the application using Netlify serverless infrastructure and developed content ingestion, offline fallback, and backup workflows.

Photon-Sim – *Photonic AI Accelerator Simulator* *September 2025 – March 2026*
Tech: *Python, NumPy, SciPy, FastAPI, React, TypeScript*
- Built a full-stack simulator with a Python computation core, FastAPI backend, and React/TypeScript dashboard to compare simulated photonic accelerator workloads against CPU baselines.
- Developed a CLI runner to automate benchmarks across matrix sizes 64–1024, exporting reproducible CSV/JSON results for accuracy, latency, and efficiency comparisons.